



999



August 26, 1998

St Vrain Community Watchdogs
Richard Cargill
5976 Hygiene Road
Longmont, CO 80503

Dear Mr. Cargill:

Recently there has been a series of letters, petitions, and phone calls regarding operating permits and concerns about ongoing compliance at our Lyons cement plant. The "St. Vrain Valley Community Watchdogs" have been very thorough and frequent in their campaign. Southdown takes these issues very seriously regardless of whether they fall into regulated areas or simply are matters of neighborhood relations.

Representatives of the Lyons plant and our Corporate staff recently met with the Watchdog group to listen to their concerns and develop action plans. We believe it was a good meeting at which we were able to discuss the issues from our neighbors; and at the same time, we were able to correct some of the rumors that had been circulated. I feel we made real progress. We opened communication channels and the Company agreed to several actions. The key ones were 1: to have a follow-up meeting in October, and 2: to distribute a newsletter explaining what we were changing and the likely impacts on the community. Southdown agreed that this would be an excellent way to address these issues. A copy of this letter is attached for your reference.

The situation started with concerns about material blowing out of an old quarry area during high wind situations. The Company addressed this issue by:

- placing cover over the old quarry area
- installing sprinklers at a new quarry area
- reducing the amount of material exposed to potential wind erosion
- reducing the haul distance for the material due to the close proximity of the new quarry area.

The Company made these changes before the date promised (June 30, 1998) and there has been no recurrence or subsequent complaints related to this area.

More recently the "St. Vrain Valley Watchdogs" have expressed concern over air permit changes requested by the Southdown. As submitted, the permit

application indicates that there would be an increase in the total tons of particulate matter emitted annually. However, the Company elected to pursue an expedited permitting process, while less time consuming, it does not allow credit for particulate matter emission reductions as a result of other plant improvements.

If these other improvements are included, the estimated emissions show a dramatic decrease in particulate matter emissions. When this information was explained to the Watchdog group at our recent meeting, they were visibly relieved. They had the misconception that particulate matter emissions would double as a result of the permit modification.

Lastly, there was discussion at the meeting about occasional fugitive emissions from the manufacturing facility during high wind conditions. Although Southdown has been judged to be in full compliance with all current permits and air emission regulations as evidenced by recent agency inspections, we are also concerned about this phenomenon. The Company is pursuing a number of equipment changes that should make improvements in this area. These items represent substantial expenditures of capital resources and will take time to implement. We are expediting efforts in this regard and anticipate that they will be in place before or during the first quarter of 1999.

Thank you for the opportunity to share these recent developments between Southdown, Inc. and the St. Vrain Valley Community Watchdogs.

Sincerely,



John W. Lohr
Plant Manager

Attachment--Newsletter (5 pages)

Southdown News

Lyons Cement Plant
5134 Ute Highway; PO Box 529
Lyons, CO 80540
(303) 823-2109

What is new at Southdown?

We are improving cost and quality; while, we reduce waste and particulate emissions. This past spring, the plant submitted an application to amend our air permits to support this work. The project consists of two major operational changes at the plant which will dramatically improve particulate emissions from the facility.

1. High Efficiency Finish Mill Separator -- The plant is currently using two obsolete separators in the finish mill circuit. The typical cement particle passes through the mill 4 or 5 times before it is ground to the required fineness. The separator is supposed to separate fine material which requires no additional grinding from the coarser particles. Unfortunately, the old separators do a poor job of this separation. As a result, a substantial amount of fine material is returned to the mill and is ground to "super-fines". This material is a bad actor in concrete batches because it requires more water to be added to achieve good workability. The higher the water addition, the lower the potential strength of the concrete.

The high efficiency will virtually eliminate these super-fines which will result in a dramatic improvement in product quality. As a consequence of this change, the mill capacity will increase slightly which will help the plant meet the summer peak demands for cement. Typically, 70+% of our product is shipped in the six months from May - October, the peak construction season.

Air movement is the principle used for separating fine and coarse particles. This high efficiency separator has a large fan and baghouse as key elements of its operation. Any new discharge to the atmosphere must be permitted. This fan and associated baghouse is a large portion of the permit changes.

2. Recycling of CKD -- We heat pulverized limestone and shale to 2700 degrees F in our kiln using coal or natural gas. Our Kiln system is equipped with large fans and baghouses to trap the particulate that is pulled through the system along with the combustion gases. The material collected in the baghouse is called "Cement Kiln Dust" (CKD). This material is very close to finished product in its chemical and physical characteristics. Unfortunately, it has a higher sulfur content than desirable. Much of this material is returned to the kiln. However, some portion cannot because of the sulfur limit for our finished

product. This material is pugmilled with water and loaded into trucks for storage in our old quarry pits.

The raw material containing most of the sulfur is our shale. The sulfur is contained in the kerogen (shale oil) impregnated in the shale. Although there is some fuel value contributed by the shale, the sulfur creates many process problems, especially the wasting of large amounts of CKD. We must purge the system of sulfur and this typically requires the elimination of over 200 tons per day from the kiln system.

We are experimenting with the reduction or elimination of this shale from our kiln mix. We are introducing a small amount of flyash and increasing our use of limestone in the kiln feed mix. This is showing promising results. It appears that we have been able to cut CKD loss in half; while, increasing production by 5 to 7%. The only negative of this program so far is the fact that it may use up our limestone supply in Dowe Flats faster than we originally planned. In addition, we must dig much deeper and move a lot more overburden in order to recover the additional limestone. This additional rock movement also requires changes to existing permits. This is the second key area we have requested permit changes for.

Permit Change Requests and Emission Netting

Boulder County is included in the metro-Denver "Non-Attainment with EPA Ambient Air Quality Standards" regarding particulate. Although the air quality is generally very good compared to these standards, there has been an historic pattern where sensors located typically in Boulder and Denver have recorded 1 or more days per year above this standard. As a consequence, Southdown or any company can only change their permits if emissions are reduced or do not increase by a significant level.

In order to make the permit changes required to support the projects outlined above, it was necessary for Southdown to demonstrate improvement or no significant increase in emissions to the State of Colorado Department of Public Health and Environment. For the types of changes we are planning the typical method is to complete a series of calculations which "model" past/current operations and compare these emissions to "future" planned emissions. A company is allowed to use its baseline from the last two years as the existing operations. The company is also allowed to use improvements to its existing and planned future operations in addition to the requested changes.

Southdown has been making several changes to operations which were used in these calculations and will result in a major reduction in emissions within the next 6 months.

- Haul truck capacity has increased from 75 ton payloads to 95 ton payloads with the introduction of a new fleet of trucks. This means fewer trips for the same tonnages.
- Haul distances have been reduced with the opening of the new quarry.
- The open acreage at the old, plant quarry reached a maximum of about 375 acres in the spring of 1997. We have already reclaimed 50 acres south of Hygiene road as dryland pasture and 55 acres as a lake. We are in the process of reclaiming another 220 acres which will be complete in the spring of 1999.
- Dust hauling and disposal has seen major changes. We have already cut the CKD waste in half over the last two years. We are working to halve it again. We have moved the disposal point closer to the plant with a 1/2 mile roundtrip compared to 2.5 mile roundtrip in past years.
- We are planning to install an underground reclaim belt to put rock into the plant system. This will eliminate about 80% of the loader work at the south end of the plant.

These are the key activities and projects used to net out the emissions calculated from the requested permit changes. **In fact, these changes will amount to a significant reduction in emissions on the order of 240 tons/year of total particulate and 120 tons/year of PM-10 (fine particulate less than 10 microns).** In order to expedite the permit changes, Southdown chose to make the permit changes concurrently. By making the changes in this fashion rather than doing the quarry first and the separator second, the company was not credited for many of the improvements in air quality it had made and could rightfully claim. As a result of not being credited for several emissions improvements, the actual permit application indicates that the plant would increase its emissions by 20 tons/year of total particulate and 10 tons/year of PM-10. The EPA considers these increase as below the significance level.

What does this mean to our neighbors?

It means that many of the old, grandfathered, unregulated emissions sources are being eliminated and new, permitted, regulated sources are taking their place. It means that the real particulate emissions from the facility will dramatically decrease by the spring of 1999. Yes, we believe there will be noticeable improvements.

As explained earlier, by reducing our CKD losses, plant production will increase by a small percentage, 5 to 7%. Our average truck traffic is less than 75 truck loads per day. With the conversion of some of our waste into finished product, the average truckloads on a given day may increase to about 79/day. Even peak shipping days would only be a small change from 140 to about 147.

Reduction of waste and emissions is right for the community and good for business. These changes will improve our community and will improve the efficiency of the plant.

Other Improvements!

The plant is working on other improvements to the operation as well. We are working on a number of projects that will improve plant appearance and much of the dust that gets raised during windy conditions.

- Twin Screw Pugmill to improve the wetting and mixing of CKD and water. This will further eliminate potential sources of fugitive dust.
- Quarry Watering System to wet our materials prior to loading and hauling activities. We have been able to totally eliminate visible dust generation by wetting materials prior to loading and dumping trucks.
- Purchase of a large commercial vacuum truck. This will enable the plant to more quickly and efficiently clean up material spills of mud and rock under conveyors.
- Purchase of a new, high efficiency, dry sweeper. This will allow us to keep our paved areas of the plant cleaner all year long. Our current sweeper is a wet sweeper which limits its application in the colder months and also leaves a thin film on the pavement even after sweeping.
- Even with the additional equipment, we are in the process of increasing our staffing to help us deal more quickly with cleanup work.
- Reclaim tunnel for production rock. This will eliminate 80% of the time currently used on the front end loader to place rock into the production process.

These changes will be costly, over \$1,000,000, and most will not be required to obtain the permit changes we are requesting. Southdown is doing them, because we believe it is the right thing to do for the community, our customers and our employees. All of these changes are moving through the approval system now and should be completed during the first quarter of 1999.

What is that noise?

We have had inquiries several times over recent years about sounds from the cement plant. There are two that receive repetitive inquiries.

Thunder? If you hear a low pitched rumble that sounds like thunder, it is most likely the rail car shaker. This is a large vibrator that is lowered onto the coal car to shake out the coal. It is typically operated between the hours of 7 am and 7 pm, Monday-Friday. It has sometimes been operated on overtime with employees starting as early as 3 am. We will restrict our hours in the future to 7 to 7 with an occasionally Saturday if needed.

Droning or Hum? If you hear this noise, it is probably the routine operation of the plant . We have many large fans which are responsible for this noise. Apparently this year with the frequent afternoon clouds and thunderstorms, we are having this sound occasionally bounce off the clouds and land some distance from the plant where it would normally not be heard. We have done noise measurements at our property line and are well below the 55 dB noise limit allowed by law. In fact the state highway noise is much louder than the plant.

CKD Disposal

The Company has implemented a number of changes concerning the storage or disposal of CKD. First the old pit has been closed and all CKD has been capped. The material is now being stored in pit just behind the plant. The amount of exposed CKD has been reduced to about 1/3 of what it was in the old pit. The CKD is watered six times per day with a commercial irrigation system. There have been no incidents of fugitive dust since the implementation of these measures in late June, 1998.

Comments, Issues and Concerns

If you have comments, issues or concerns regarding plant operation of conditions, please let us know. We can only impact situations that we know about. It is really helpful if people can be specific about times, observations, locations, etc. Please let us know.

Control Room: 823-2131 (24 hours per day, 7 days per week)

John Lohr, Plant Manager: 823-2109 (office hours on weekdays)

Steve Mossberg, Plant Compliance Manager: 823-2113 (office hours on weekdays)

From:

Southdown, Inc.

Lyons Cement Plant

PO Box 529

Lyons, CO 80540